DSSG FINAL - RURAL E-MENTORING

Understanding Online Mentoring Relationships

Using NLP Techniques for Analysis and Program Improvement

TABLE OF CONTENTS

THE TEAM

Makafui Amouzouvi

Tiffany Chu

Jonah Curl

02

OUR

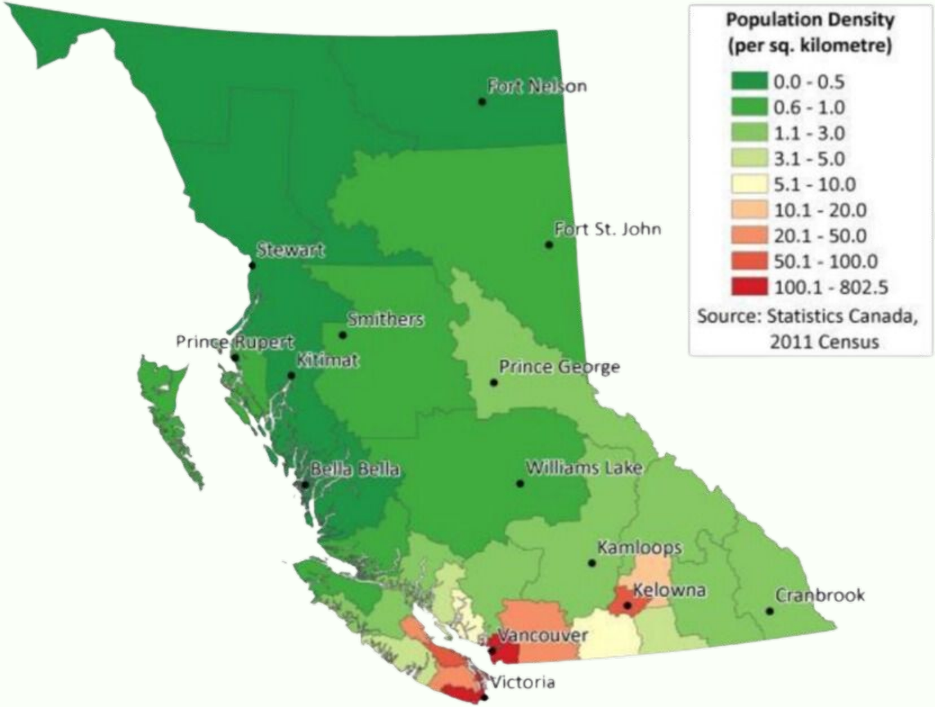
PARTNER

RURAL EMENTORING BC: MOTIVATIONS

- Rural and Indigenous Communities have **higher demand for healthcare** but reduced access.

- Rural and Indigenous Students are more **likely to find careers in their communities**.

- UBC has programs designed to improve Rural and Indigenous admissions into Health Science

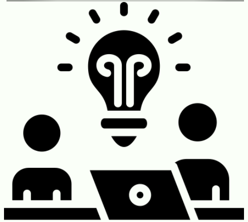


RURAL EMENTORING BC: MISSION AND

ReMBC's goal is to support and inspire new rural and indigenous healthcare practitioners in British Columbia.

- **Mentorship relationships:** one-on-one relationships through MentorCity's online mentoring (eMentoring) platform

- **Mentors:** university students in professional programs trained to support mentees through their career exploration



- **High School Curriculum:** to stimulate relationship development and conversation in various topics

03

PROJECT INTRO & GOALS

PROJECT INTRO & GOALS

Main Goal/Question :

What are successful pairs doing differently that ReMBC could help facilitate in the future?

DATASET

- Access to all text conversations in eMentoring pairs via MentorCity

- Discussions within units/categories of the curriculum including:
  - Rural to Urban
  - Career Exploration
  - “Adulting”

Response Datetime	Relationship ID	Mentor	Response	Category
2024-01-01 10:03	1584937	Mentor	Hi, Hope you are well	Posts in Ways of Knowing
2024-01-01 12:00	1584937	Mentee	I am well, thank you	Posts in Ways of Knowing

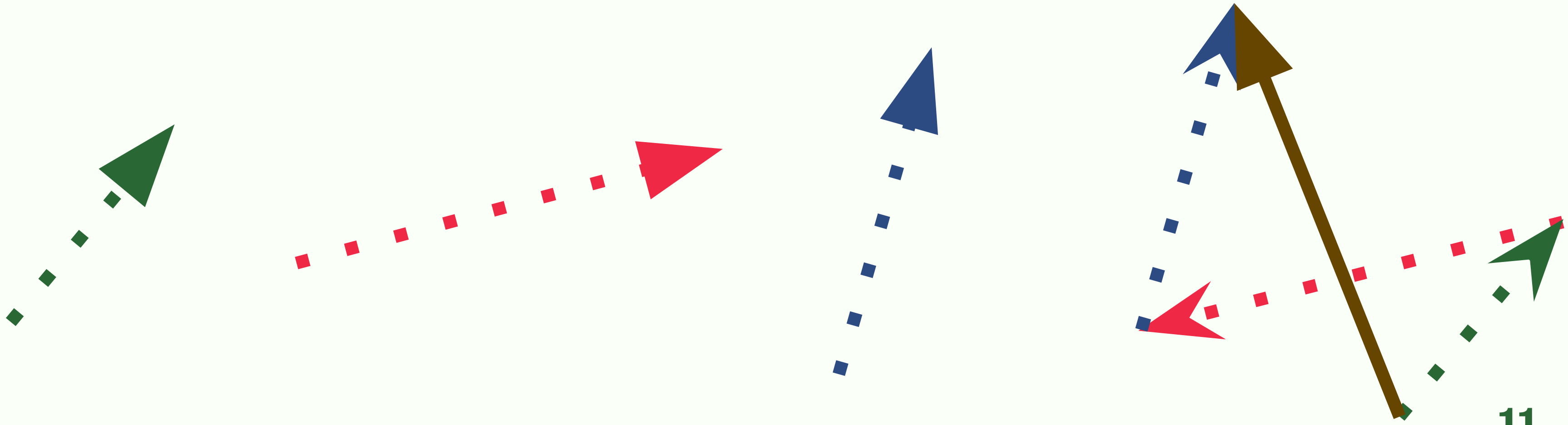
04

PROCESS & RESULTS

WORD EMBEDDINGS

Word Embeddings: words converted to multidimensional vectors, allowing for a measure of similarity using distance or angles.

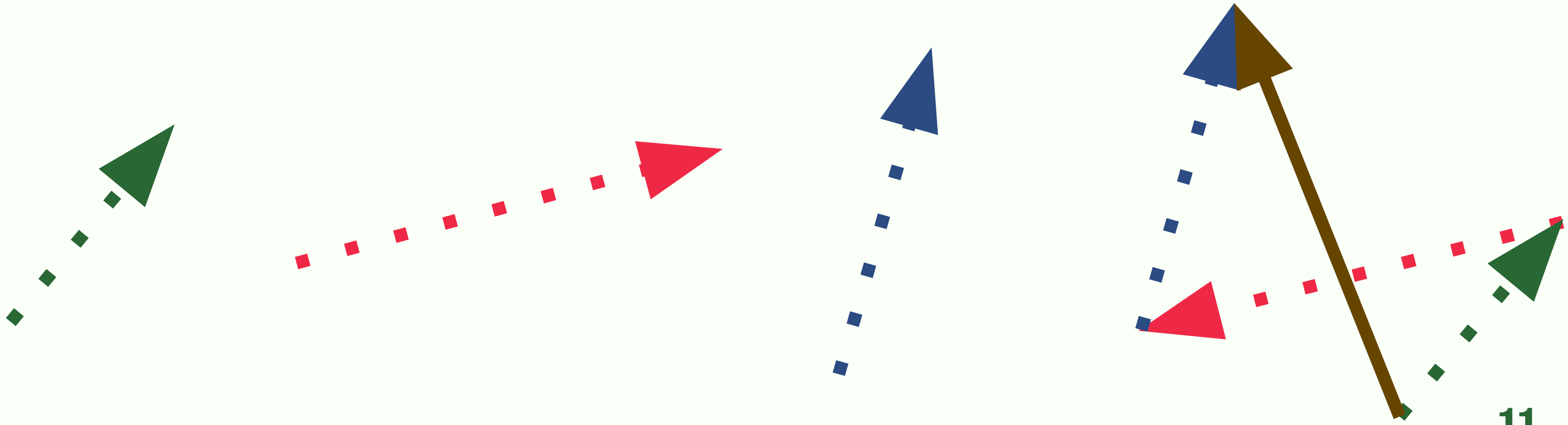
Kitten - Cat + Dog = ?



WORD EMBEDDINGS

Word Embeddings: words converted to multidimensional vectors, allowing for a measure of similarity using distance or angles.

Kitten - Cat + Dog = Puppy



1. TOPIC MODELING

Extracting common themes in conversations between mentees and mentors

BERTopic

Modular

- More customizable to specific needs

Bidirectional Embeddings (BERT)

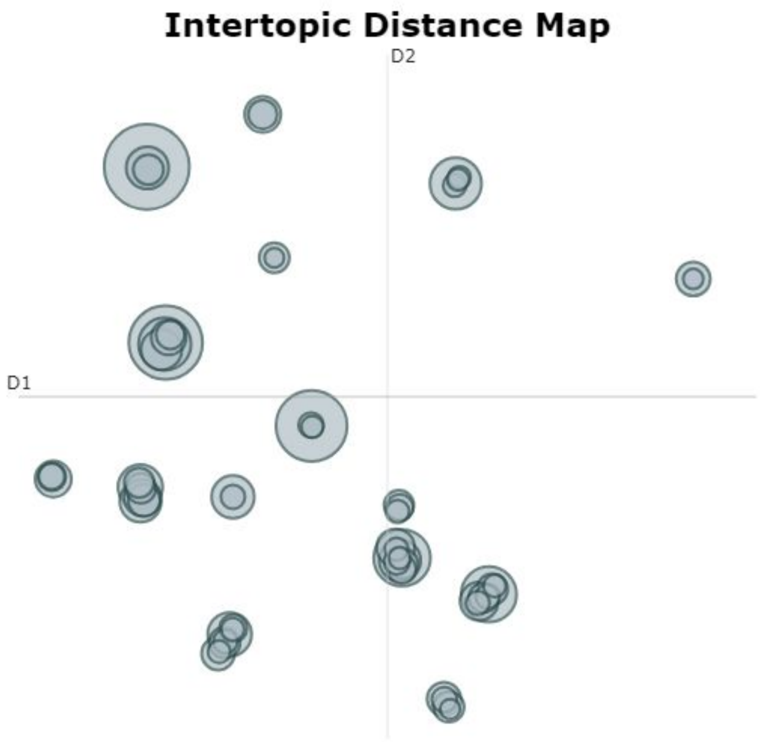
- Words are encoded based on both meaning and context in the document.

I am feeling blue. vs The water is blue.

blue =/= blue

1. TOPIC MODELING: RESULTS

- Over 60 topics from mentee responses.
- Later topics have less responses in them.



1. TOPIC MODELING: RESULTS

BERTopic

Top 10 Topics From Mentee Responses:	Representative Words:
1. **Career and Post-Secondary Exploration**	careers, college, university
2. **Rural to Urban Living**	rural, towns, community
3. **Wellness and Self-care**	health, selfcare, stress
4. **Developing Good Study Habits**	studying, memorize, practice
5. **Mentorship Conversations**	mentoring, talk, helping
6. **Free Time**	spring, holidays, break
7. **Finding Inspiration**	inspiration, passion, creation
8. **Hobbies**	hobbies, sports, skiing
9. **Online Platform**	login, messages, app
10. **Career Exploration**	careers, pursuing, profession

1. TOPIC MODELING: RESULTS

Top 10 Topics From Mentee Responses:	Representative Words:
1. **Career and Post-Secondary Exploration**	careers, college, university
2. **Rural to Urban Living**	rural, towns, community
3. **Wellness and Self-care**	health, selfcare, stress
4. **Developing Good Study Habits**	studying, memorize, practice
5. Mentorship Conversations	
6. Free Time	
7. **Finding Inspiration**	inspiration, passion, creation
8. Hobbies	
9. Online Platform	

1. TOPIC MODELING: RESULTS

Top 10 Topics From Mentee Responses:	Representative Words:
1. Career and Post-Secondary Exploration	
2. Rural to Urban Living	
3. Wellness and Self-care	
4. Developing Good Study Habits	
5. **Mentorship Conversations**	mentoring, talk, helping
6. Free Time	
7. Finding Inspiration	
8. Hobbies	
9. **Online Platform**	login, messages, app

1. TOPIC MODELING: RESULTS

Top 10 Topics From Mentee Responses:	Representative Words:
1. Career and Post-Secondary Exploration	
2. Rural to Urban Living	
3. Wellness and Self-care	
4. Developing Good Study Habits	
5. Mentorship Conversations	
6. **Free Time**	spring, holidays, break
7. Finding Inspiration	
8. **Hobbies**	hobbies, sports, skiing

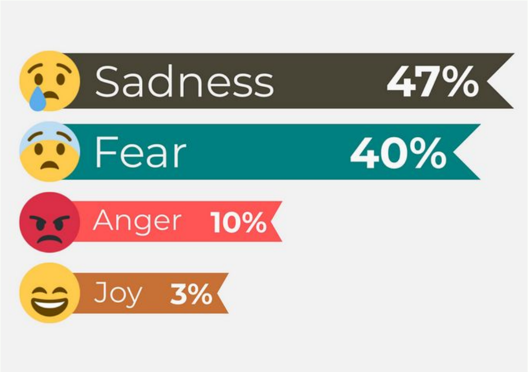
2. EMOTION DETECTION : HOW DOES IT WORK?

Method:

ZERO-SHOT CLASSIFICATION

FACEBOOK AI

+ **BART** →

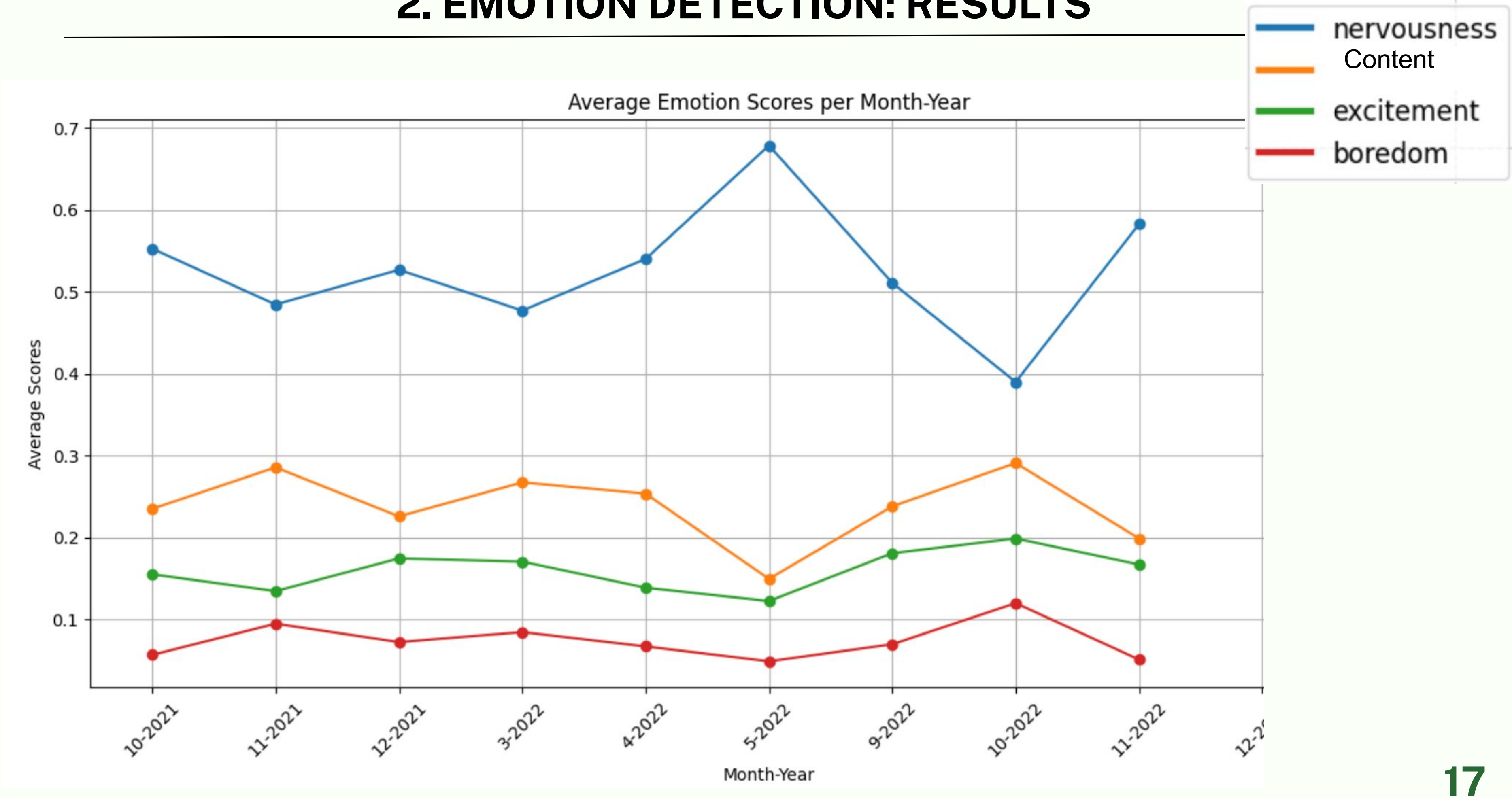


BART = **Bidirectional** and **Auto-Regressive**

2. EMOTION DETECTION: RESULTS

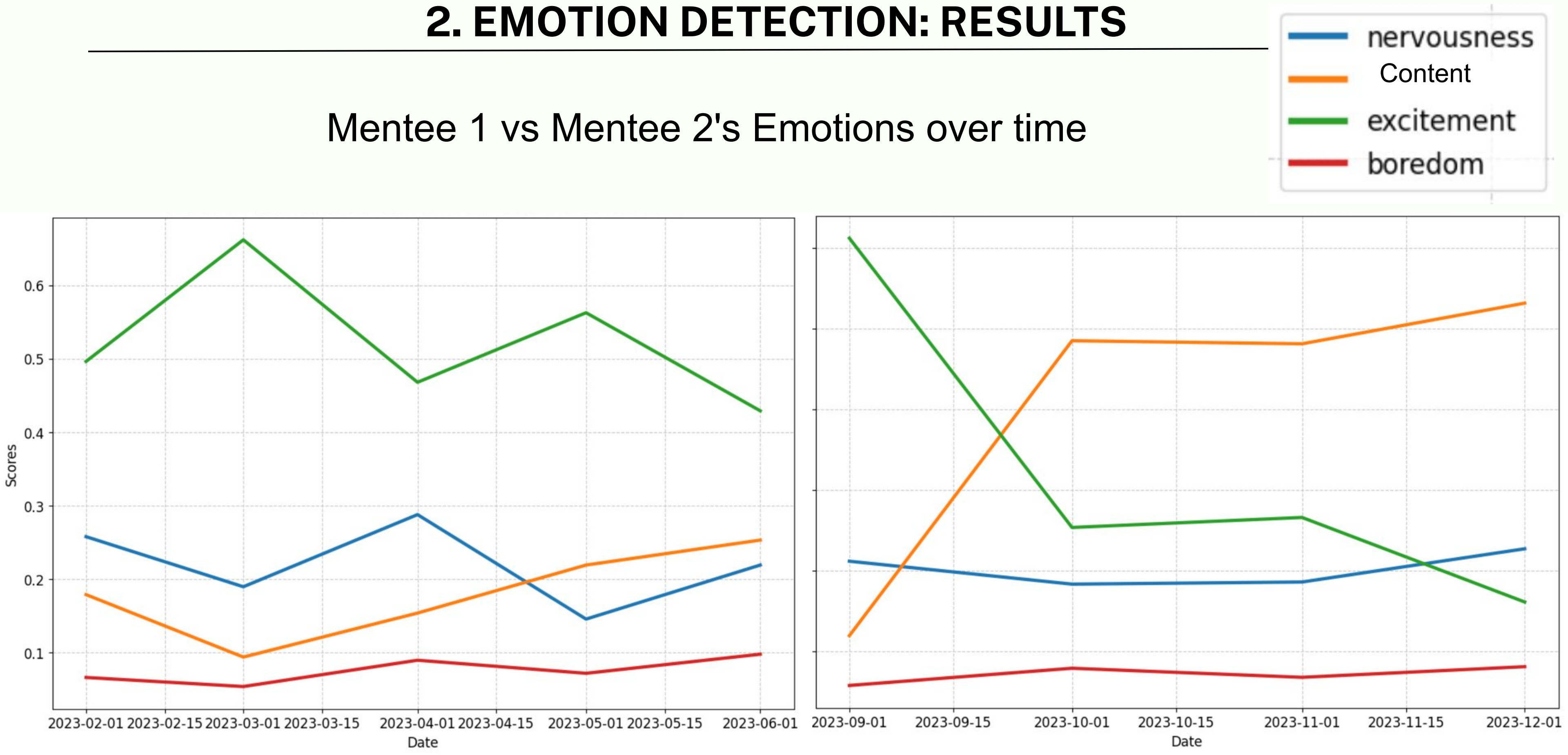
Datetime	Response	Predicted Labels	Scores
2021-09-14 16:41:00	Hey Mentor, I'm Mentee, bioengineering, wow. What do you enjoy about outdoor activities? My dad got me into them too, but skiing is definitely my favourite. **I'm looking forward to getting to know you**. I'm thinking we start with career considerations.	[excited, happy]	[0.342, 0.297]
2021-09-14 16:48:00	hi Mentor **I'm a little shy** and **have a hard time and writing down what I am thinking**. I'd like to start our conversation with choosing a program and school.	[nervous]	[0.857]

2. EMOTION DETECTION: RESULTS



2. EMOTION DETECTION: RESULTS

Mentee 1 vs Mentee 2's Emotions over time

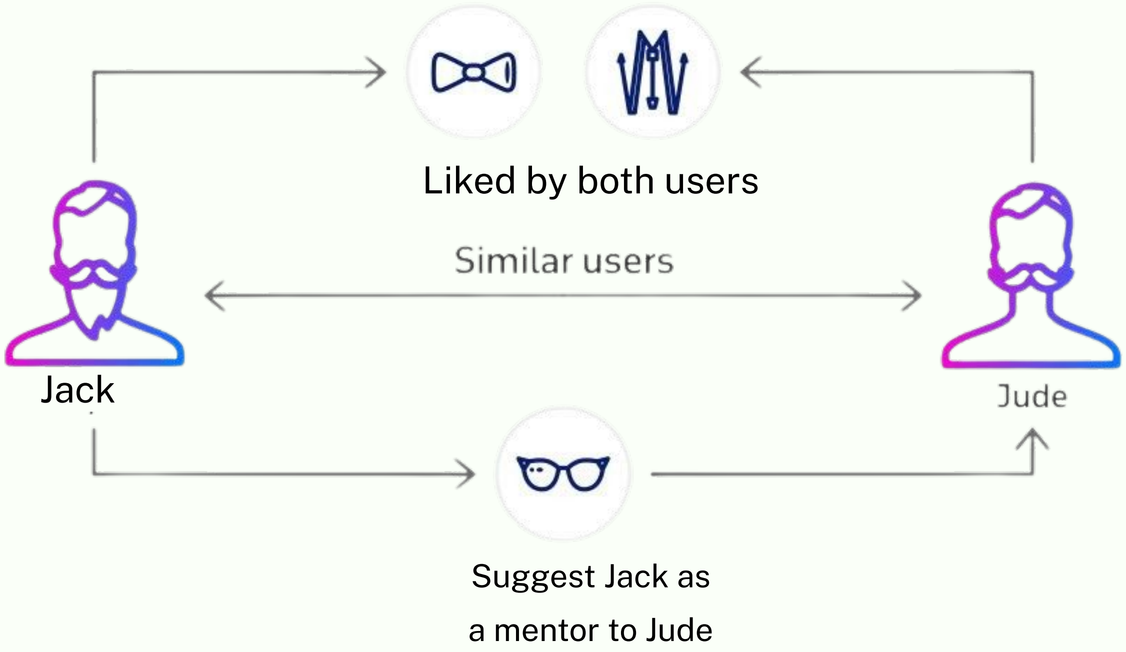


3.MENTOR-MENTEE MATCHING

Current process

- Matching based on profile surveys
- Intensive Manual work

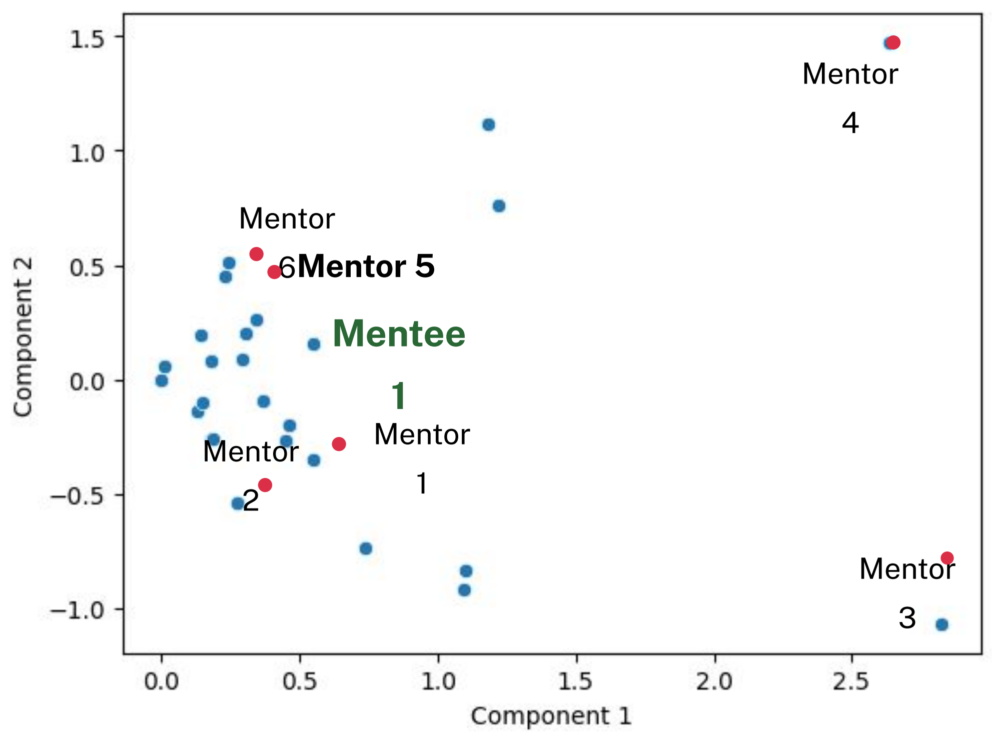
Desired process



3.MENTOR-MENTEE MATCHING RESULTS

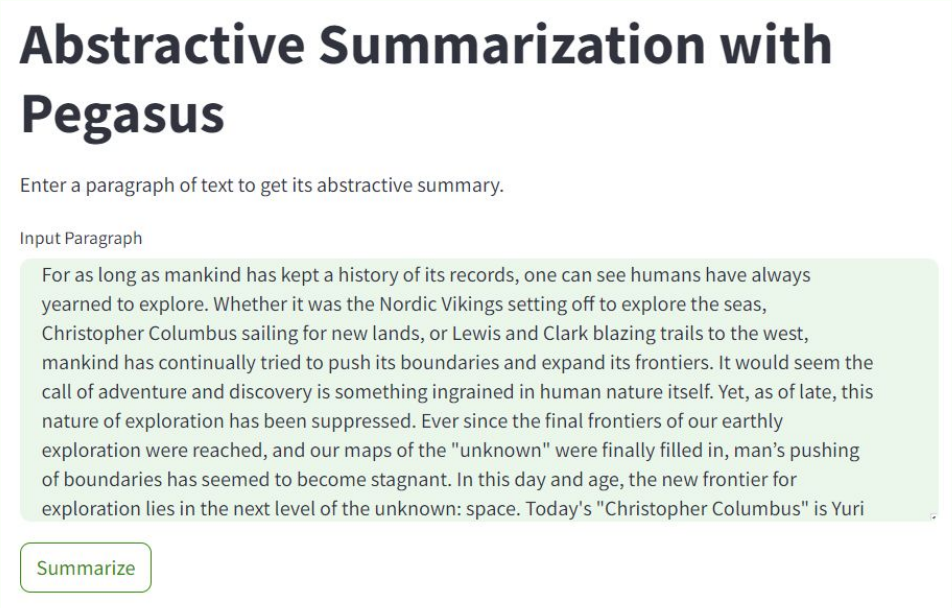
Person	Role	Nearest Neighbours
Bob	Mentor	[(Mary, 0.9, mentor), (Jane, 0.8, mentee) ...]
Jane	Mentee	[(Jack, 0.8, mentee), (Bob, 0.8, mentor) ...]

Outpu

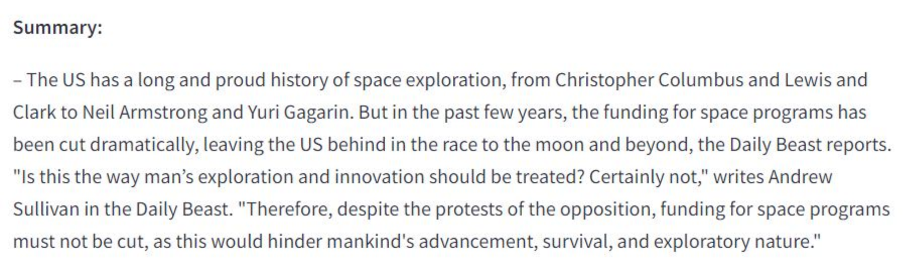


Vector space

4. ABSTRACTIVE SUMMARIES

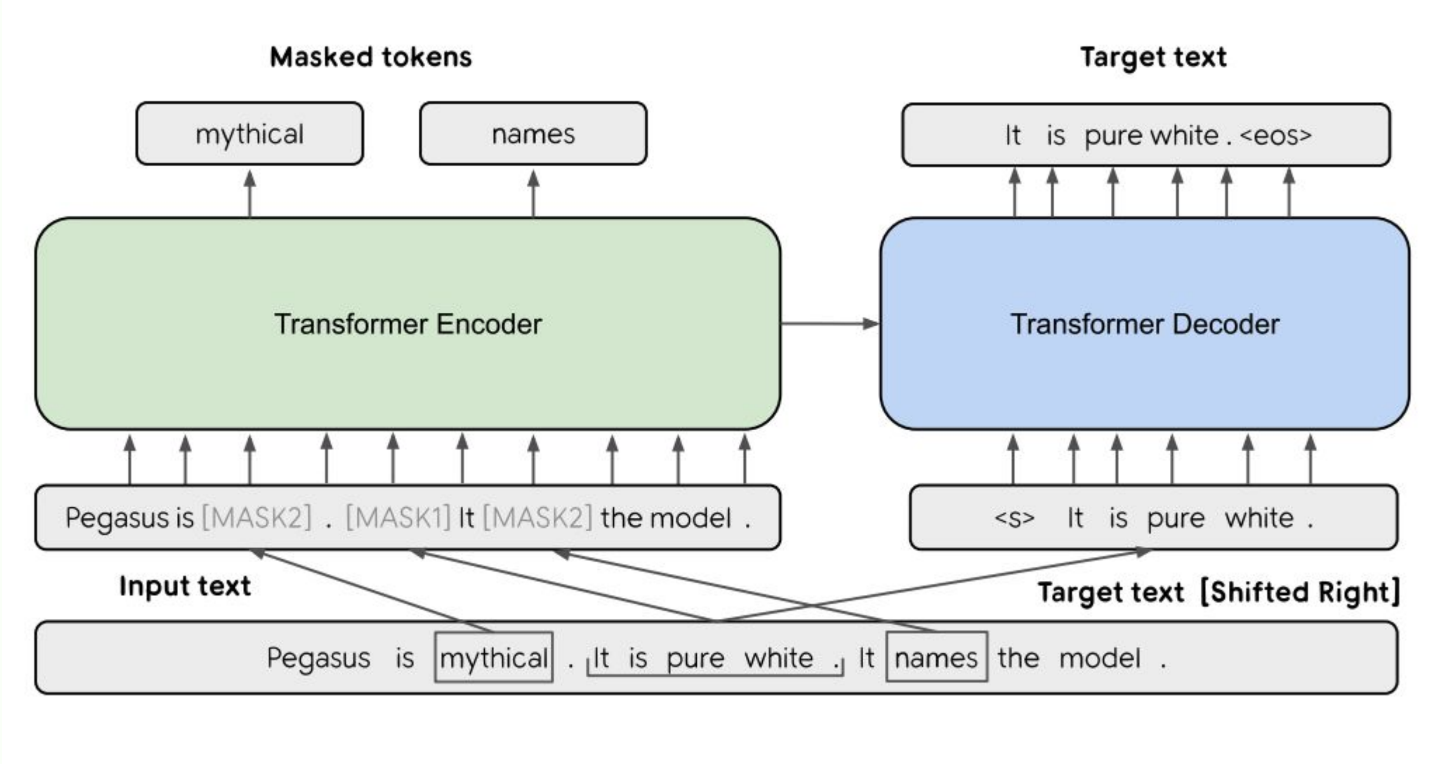


example input paragraph



output from our app

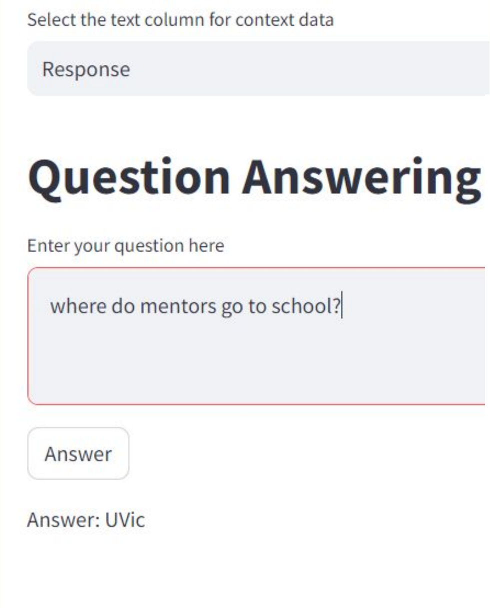
4. ABSTRACTIVE SUMMARIES

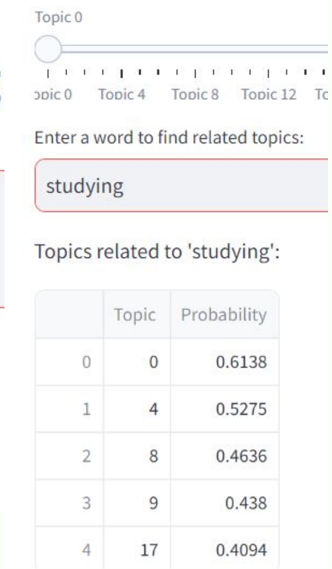


Pegasus Transformer showing Gap Sentences Generation (GSG)

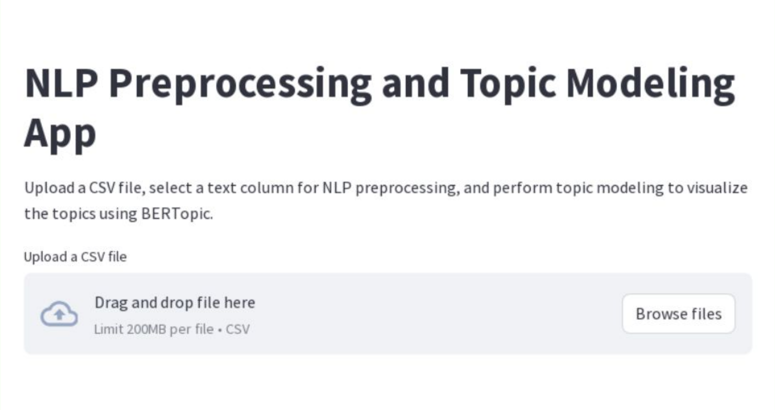
5.DEPLOYING A NLP APPLICATION

This app allows researchers to upload their own files to use NLP on their own data without doing any of the coding themselves



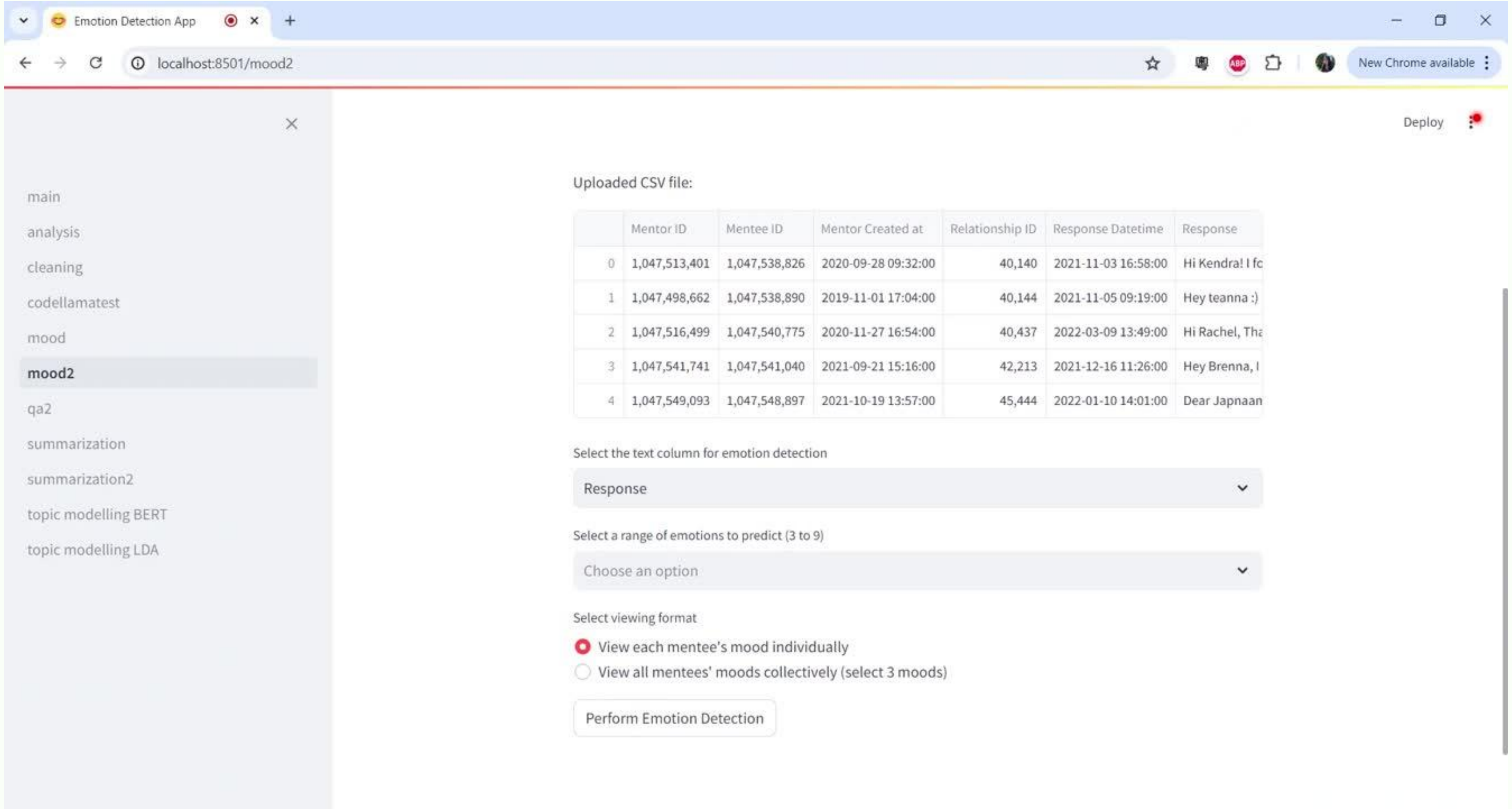


query

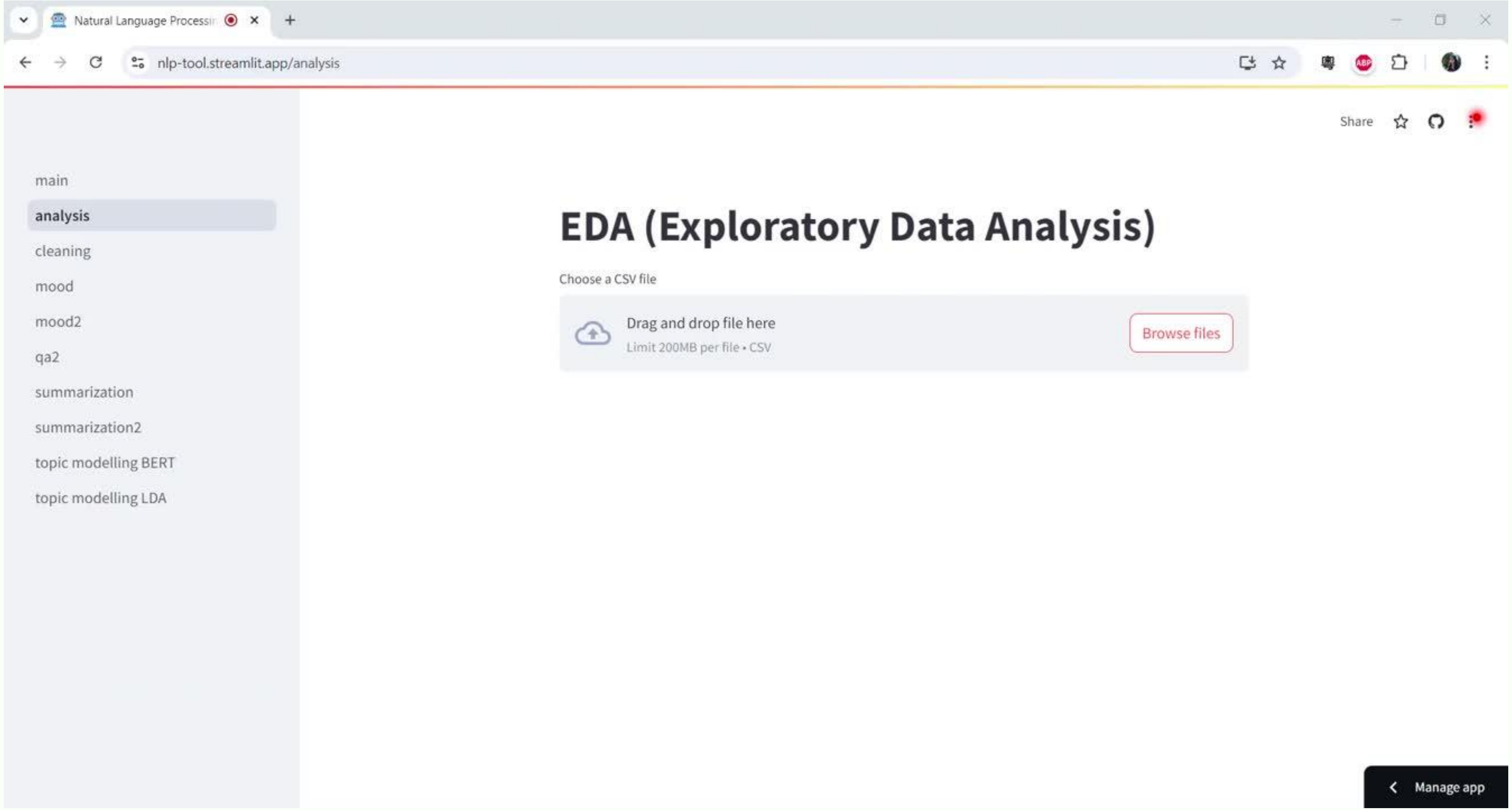


csv upload

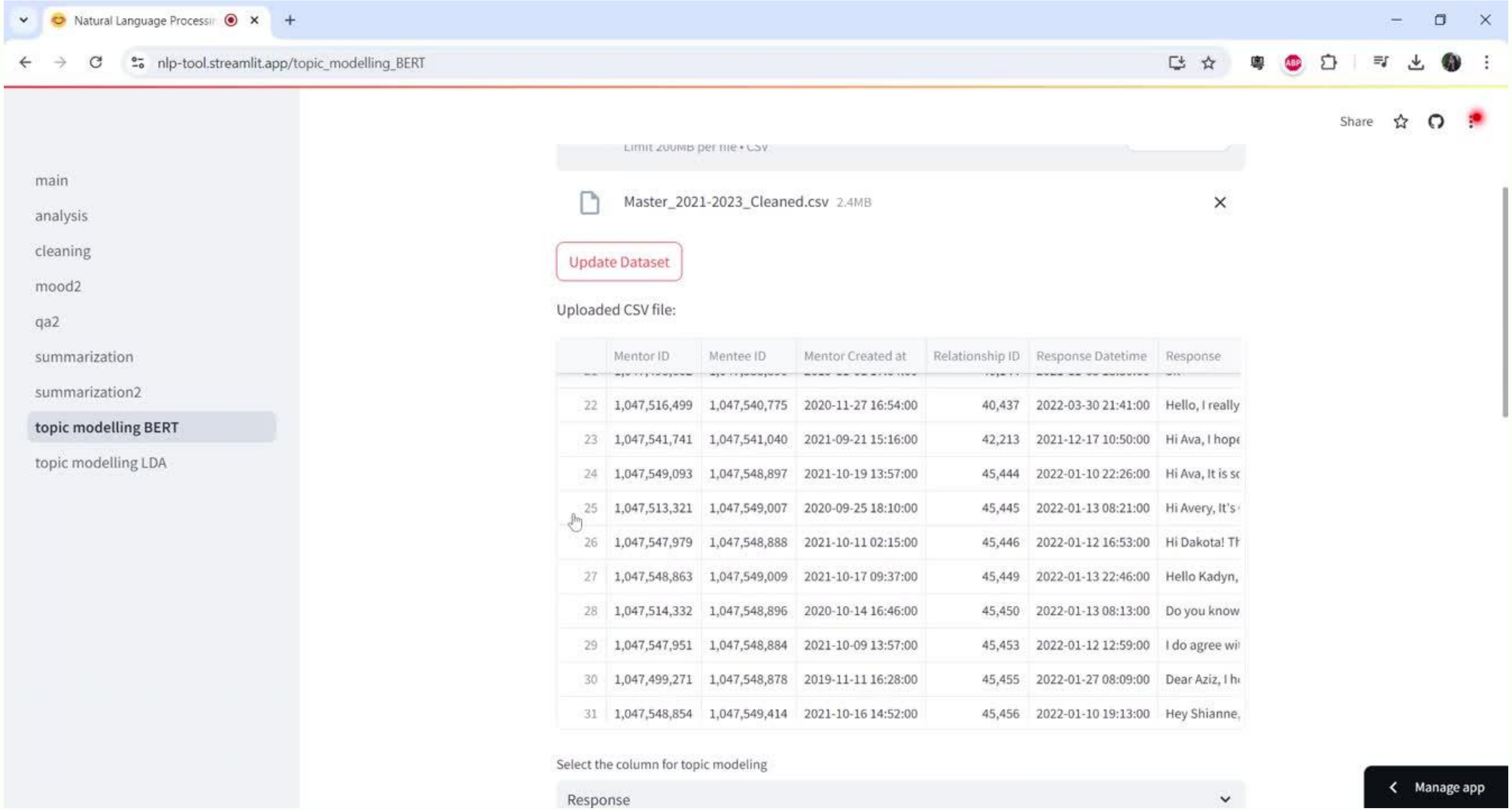
EMOTION DETECTION DEMO



ANALYSIS AND FILTERING DEMO



BERT TOPIC MODELLING DEMO



	Mentor ID	Mentee ID	Mentor Created at	Relationship ID	Response Datetime	Response
22	1,047,516,499	1,047,540,775	2020-11-27 16:54:00	40,437	2022-03-30 21:41:00	Hello, I really
23	1,047,541,741	1,047,541,040	2021-09-21 15:16:00	42,213	2021-12-17 10:50:00	Hi Ava, I hope
24	1,047,549,093	1,047,548,897	2021-10-19 13:57:00	45,444	2022-01-10 22:26:00	Hi Ava, It is sc
25	1,047,513,321	1,047,549,007	2020-09-25 18:10:00	45,445	2022-01-13 08:21:00	Hi Avery, It's
26	1,047,547,979	1,047,548,888	2021-10-11 02:15:00	45,446	2022-01-12 16:53:00	Hi Dakota! Th
27	1,047,548,863	1,047,549,009	2021-10-17 09:37:00	45,449	2022-01-13 22:46:00	Hello Kadyn,
28	1,047,514,332	1,047,548,896	2020-10-14 16:46:00	45,450	2022-01-13 08:13:00	Do you know
29	1,047,547,951	1,047,548,884	2021-10-09 13:57:00	45,453	2022-01-12 12:59:00	I do agree wi
30	1,047,499,271	1,047,548,878	2019-11-11 16:28:00	45,455	2022-01-27 08:09:00	Dear Aziz, I h
31	1,047,548,854	1,047,549,414	2021-10-16 14:52:00	45,456	2022-01-10 19:13:00	Hey Shianne,

05

RECOMMENDATIONS

&

FUTURE DIRECTIONS

RECOMMENDATIONS & FUTURE DIRECTIONS

Label relationships successful or unsuccessful. Can introduce 3+ labels (e.g. most engaged, engaged, not engaged)

LABEL DATA

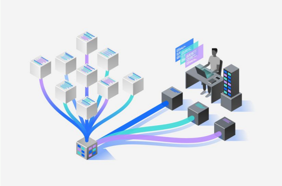
Apply tone detection to mentor and measure student engagement accordingly. How do different tones and writing styles affect engagement ?

TONE DETECTION

05

CONCLUSION

CONCLUSION

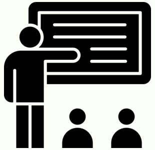


topic modeling

summarization

mood detection

university guidance



curriculum improvement & training

ANY QUESTIONS?

THANKS!